a map of meaning

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1 a map by hand
2 tokens get numbers
3 computing with meaning
4 from tokens to a text
5 the model's map
6 what the map does not know
7 inside the numbers

before a computer can search for meaning, it has to know where each text belongs.

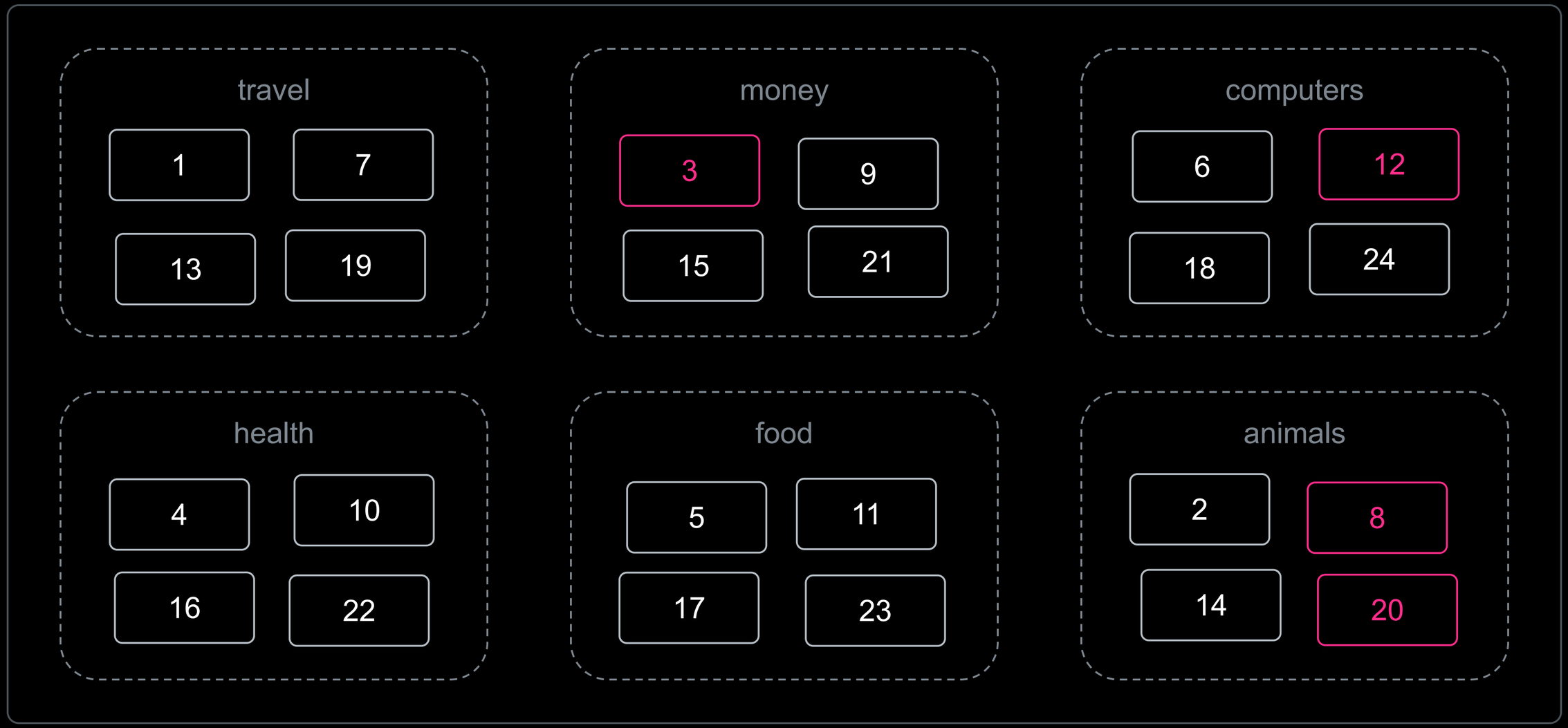


image: ai-generated (gpt-image-2.5-flare)

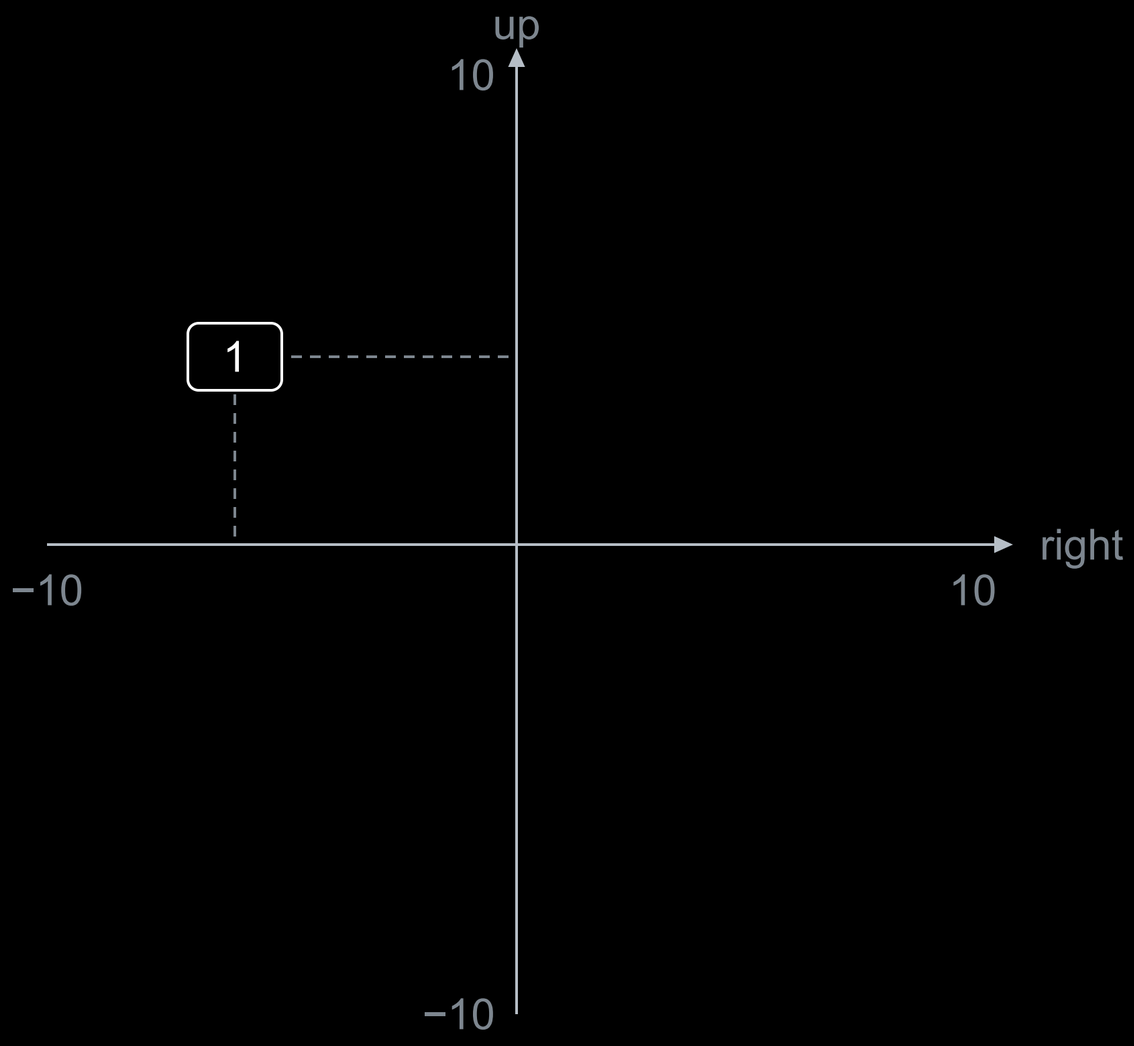
part 1

a map by hand

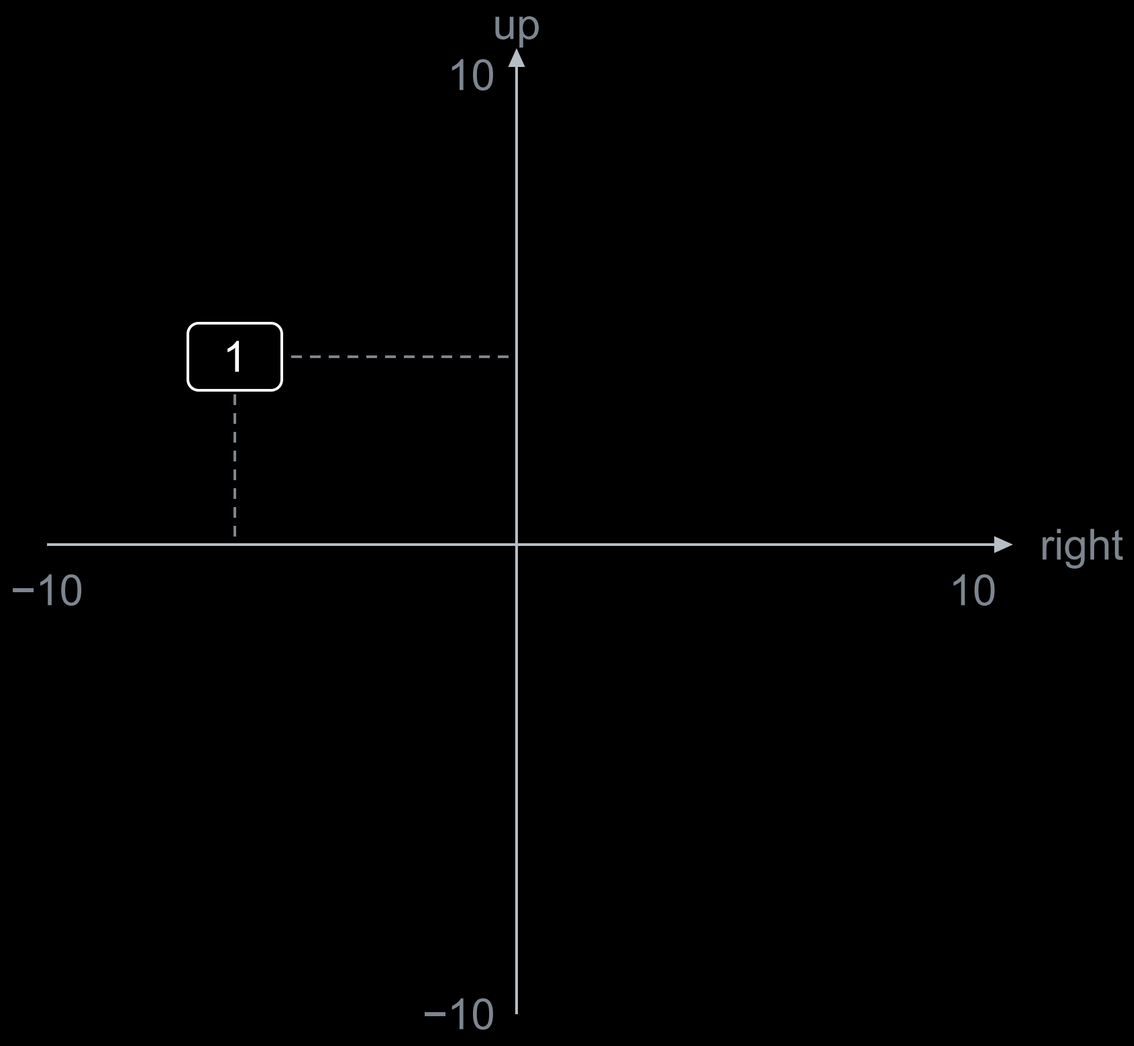
close in meaning, close on the map



a place is a list of numbers

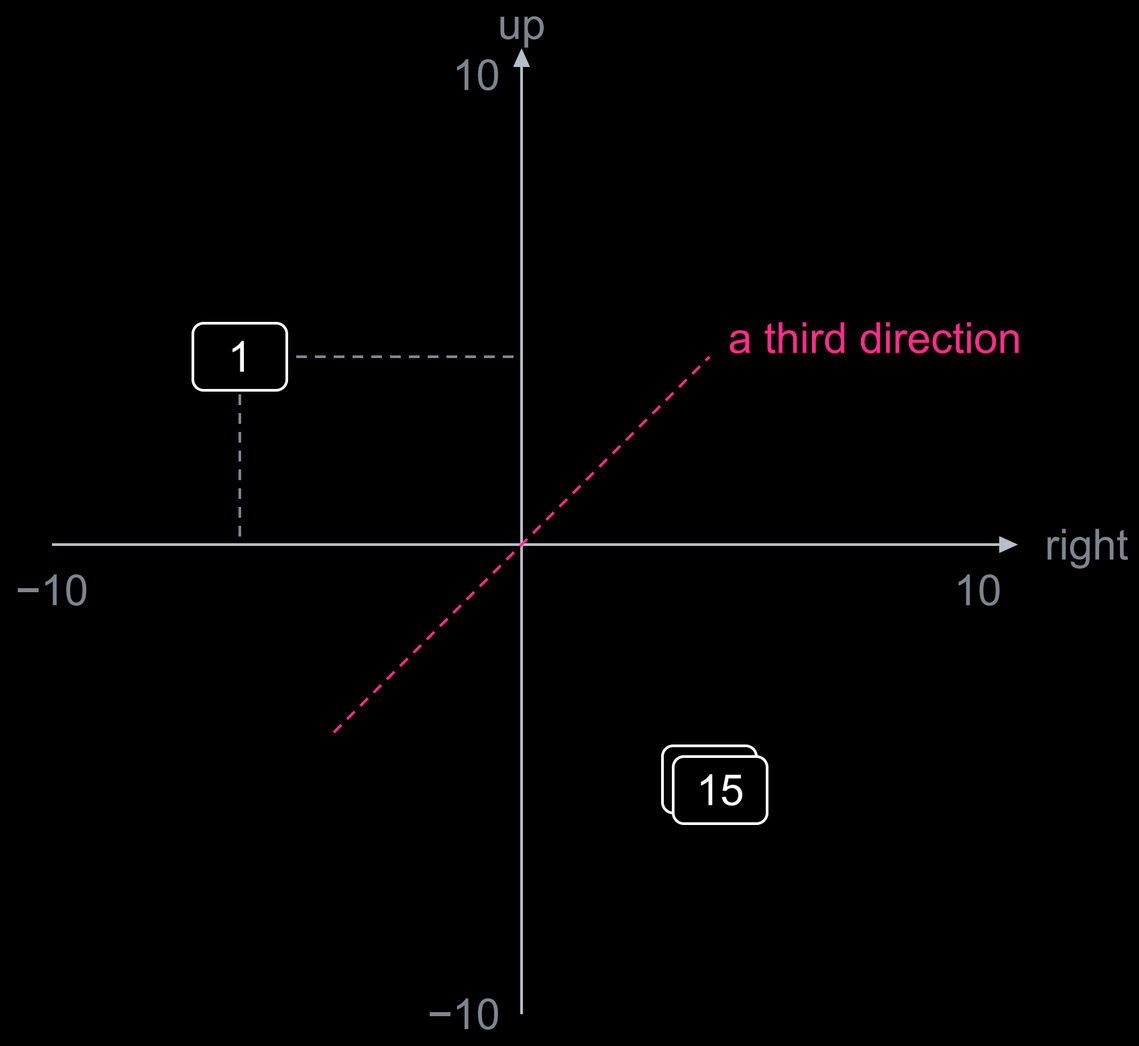


a place is a list of numbers




```
card 1 → (-6, 4)
```

a place is a list of numbers




```
card 1 → (-6, 4)
```

two cards on the same spot:

```
card 9  → (4, -5, 2)
card 15 → (4, -5, -6)
```

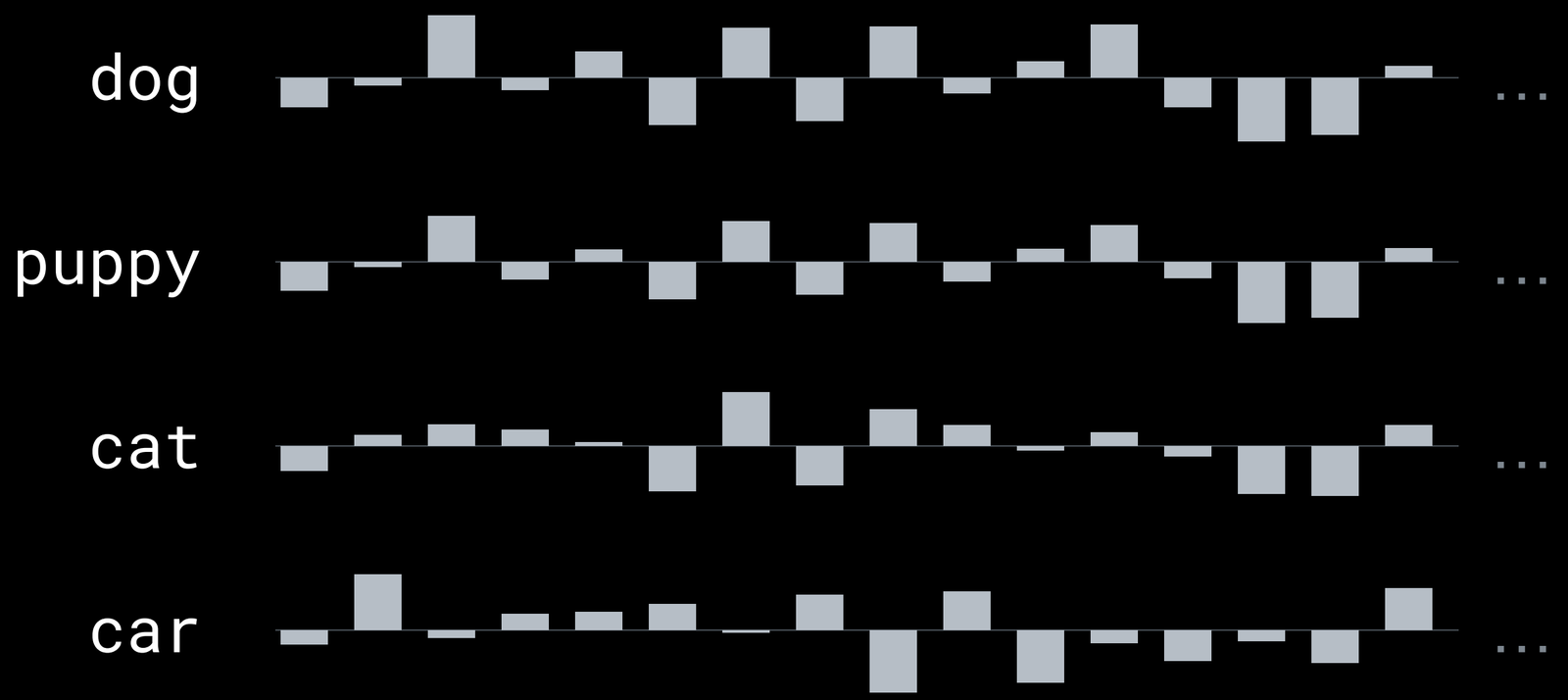
an embedding is a list of numbers that says where something lies on a map of meaning.

your sheet: 2 numbers per card · the model: 384 numbers per token and per text

part 2

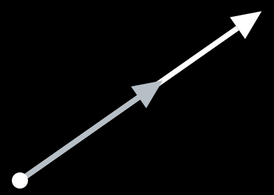
tokens get numbers

a table with one row per token



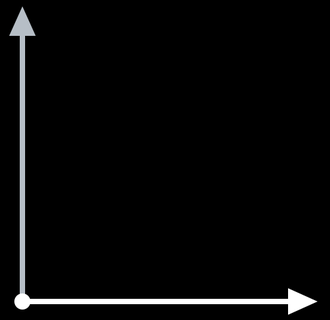
30,522 tokens × 384 numbers

similarity is the angle between two arrows



1

same direction



0

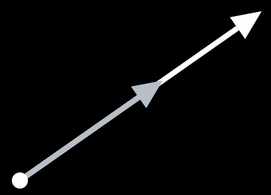
at a right angle



−1

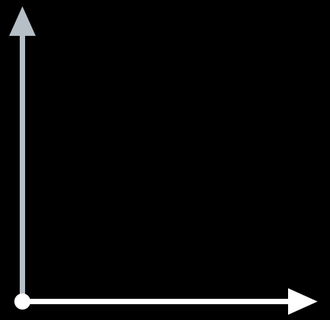
opposite

similarity is the angle between two arrows



1

same direction



0

at a right angle

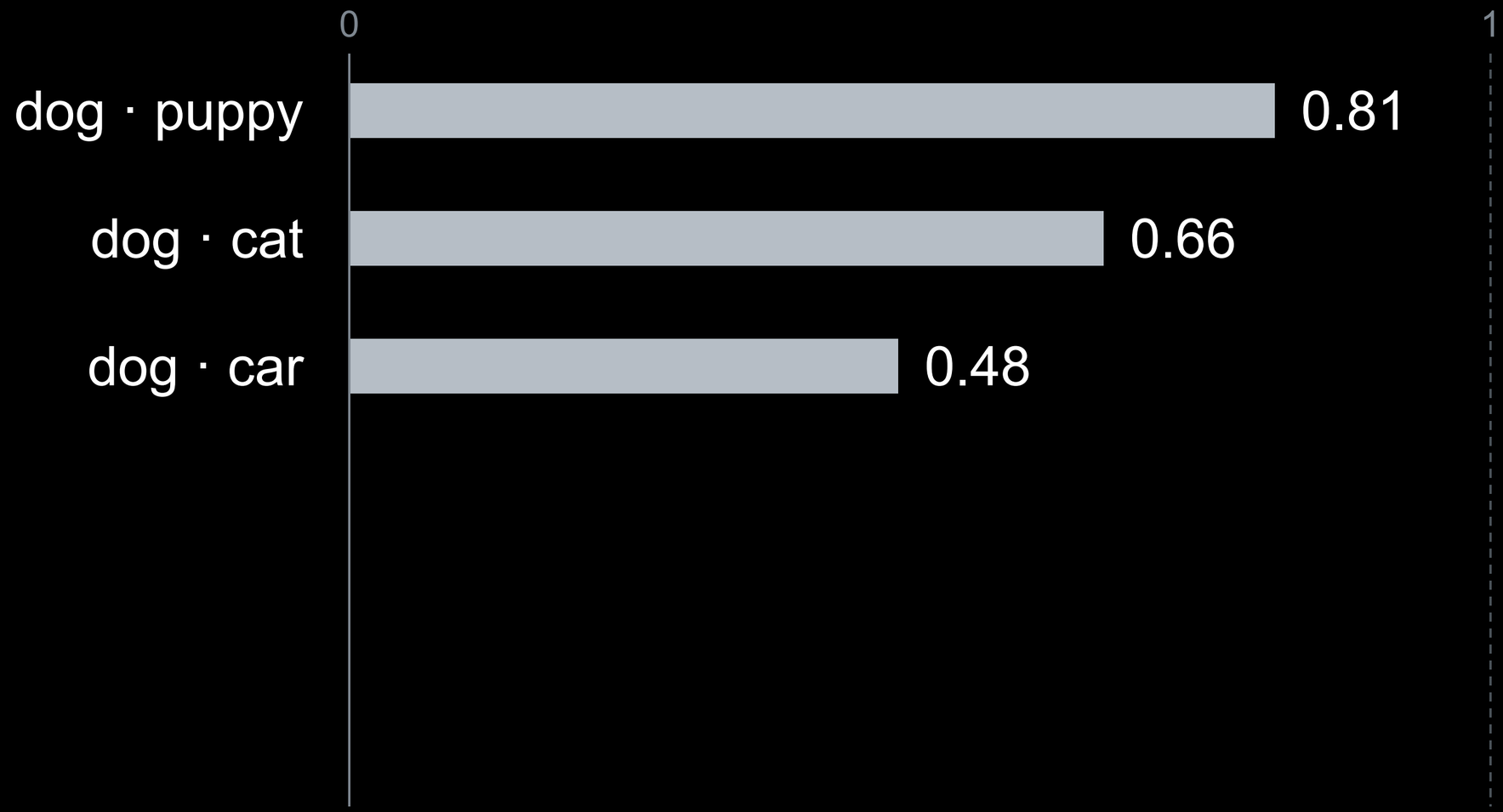


−1

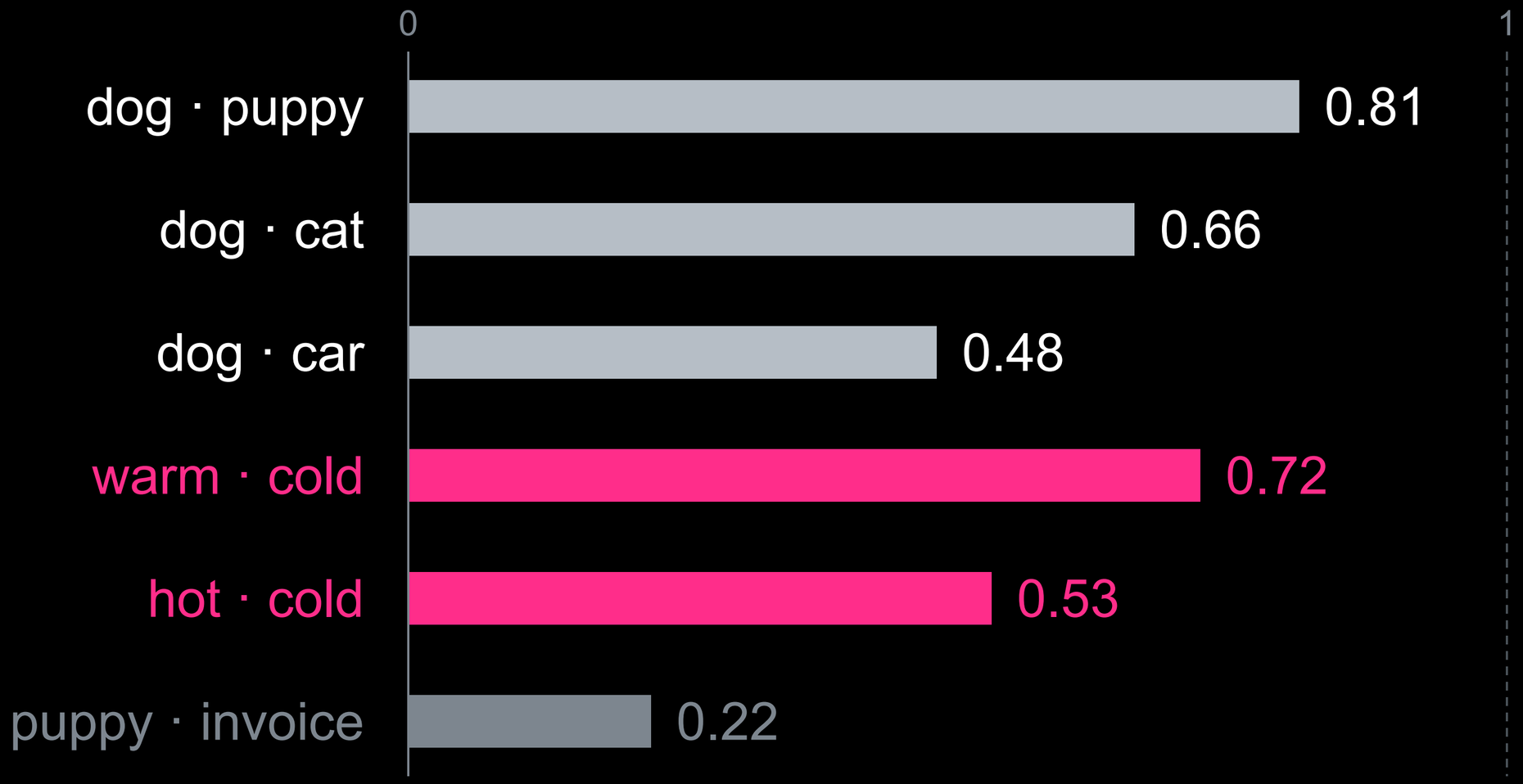
opposite

almost never: the lowest of
1.3 million word pairs is −0.09

measured with the model



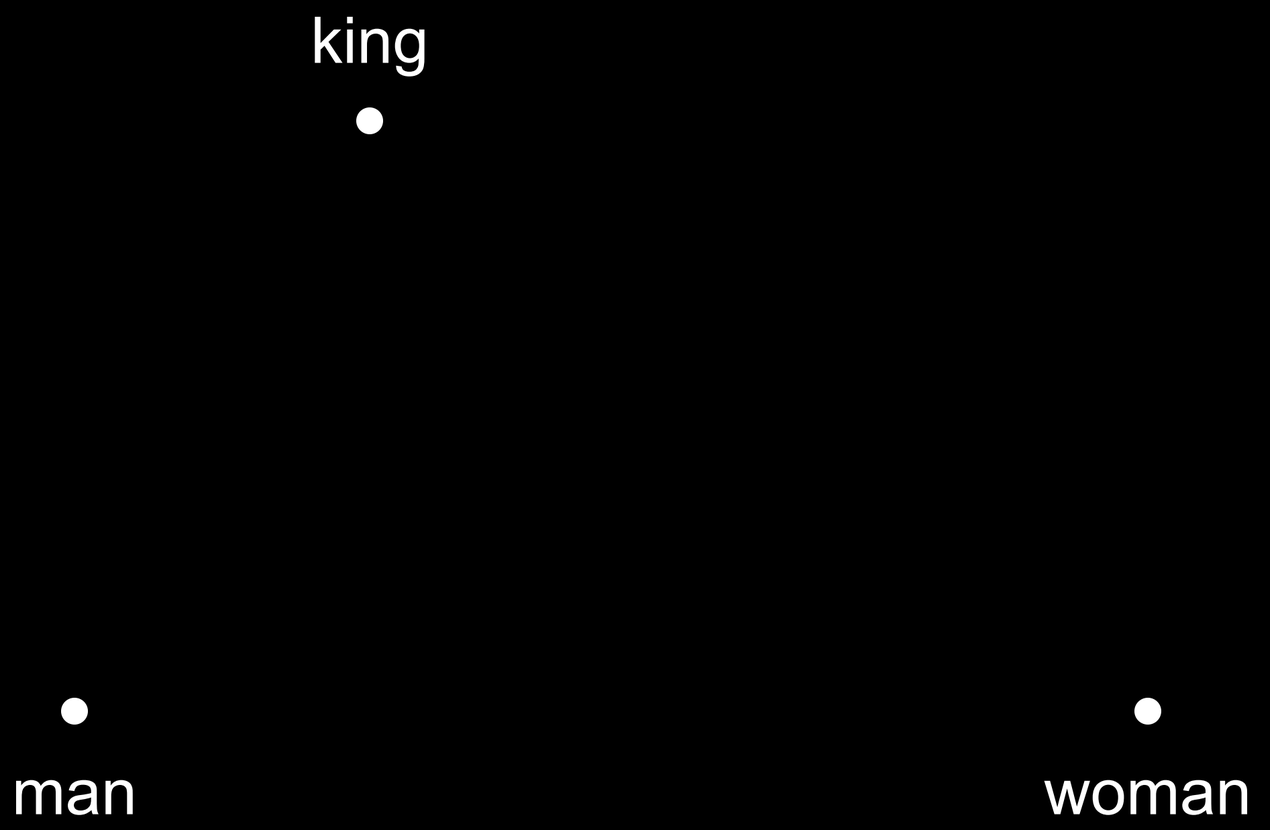
measured with the model



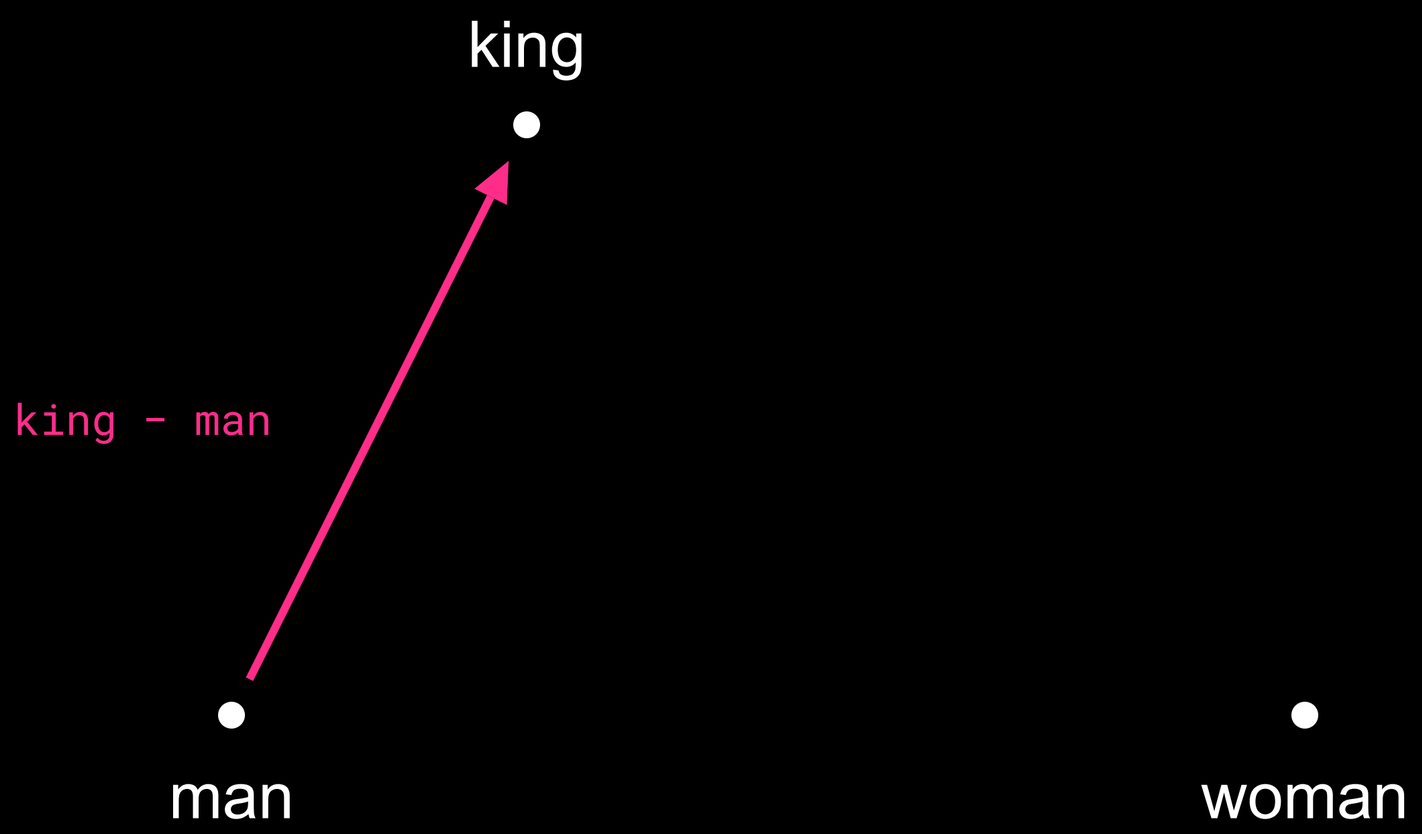
part 3

computing with meaning

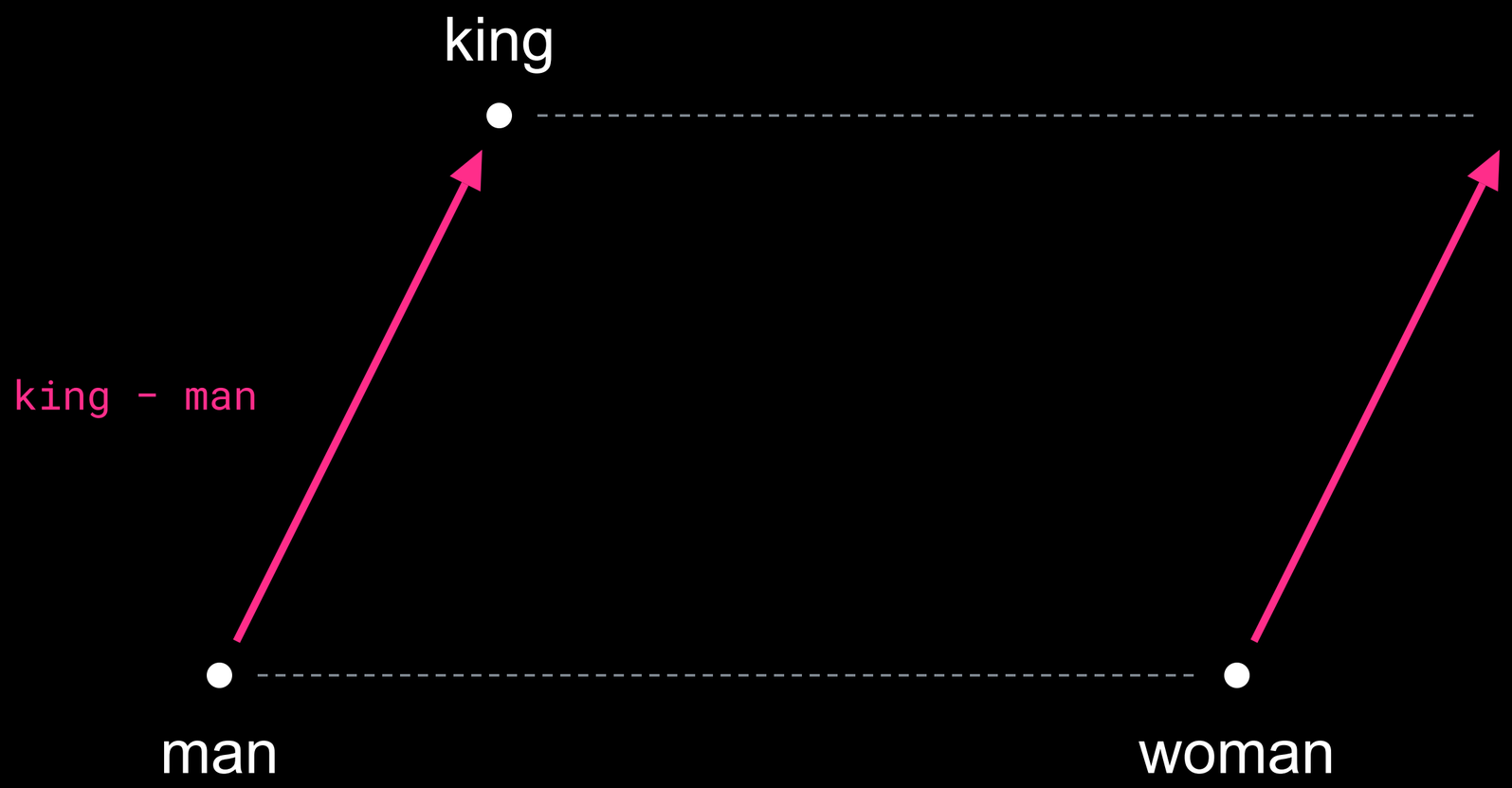
king − man + woman



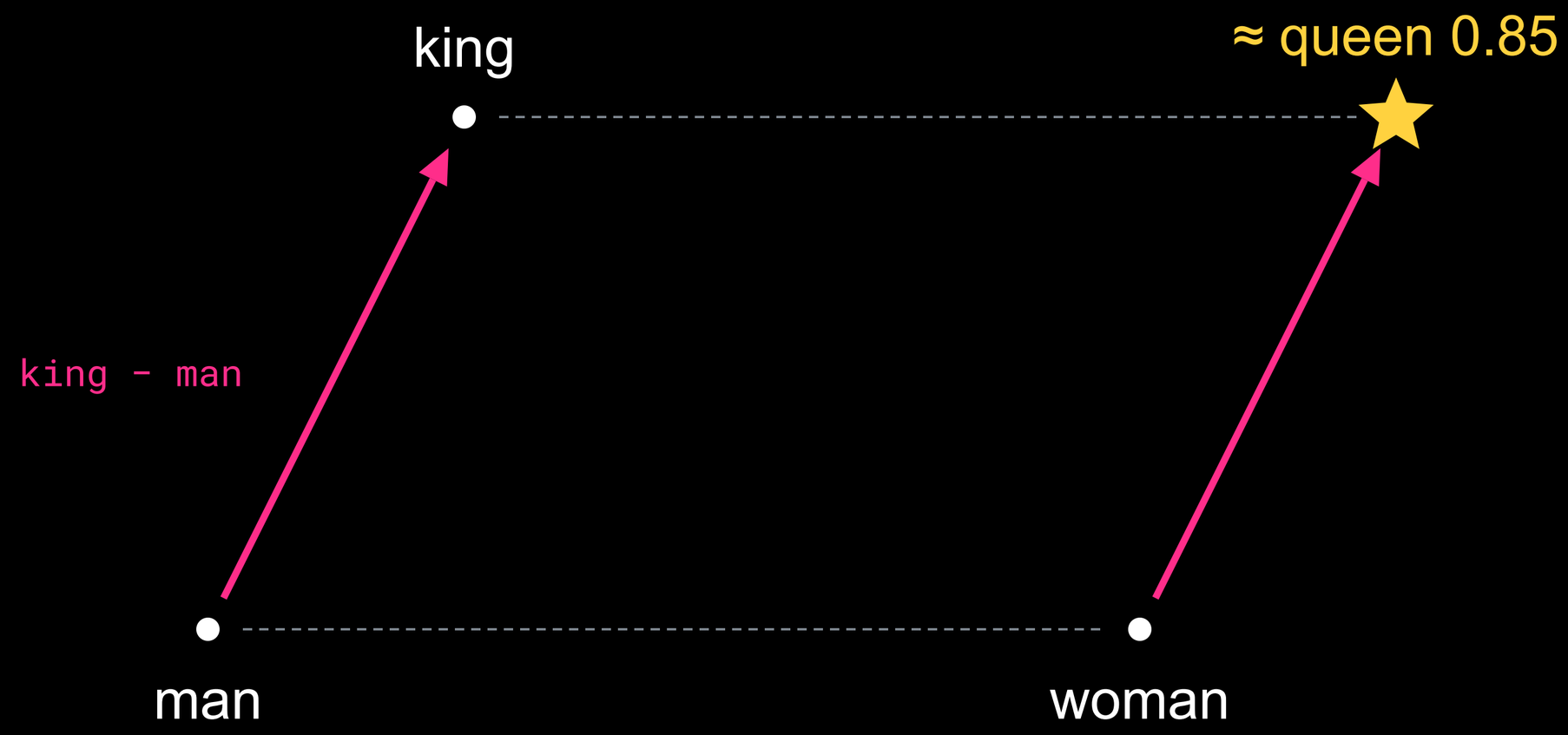
a sketch, not to scale



a sketch, not to scale



a sketch, not to scale



a sketch, not to scale

```
paris - france + germany    → berlin 0.79
puppy - dog + cat           → kitten 0.87
doctor - hospital + school  → teacher 0.82
walked - walk + swim        → swimming 0.91 swam 0.88
```

```
queen - woman + man
```



part 4

from tokens to a text

I can't get home tonight



unbelievably



ChatGPT



Ich bin krank



unbelievably · incredibly 0.35

four steps from tokens to one list

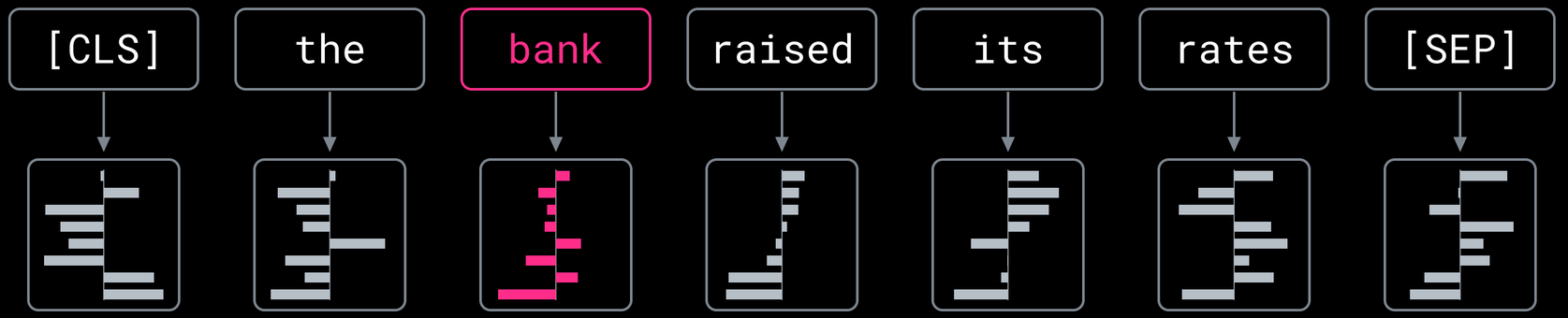
cutting

[CLS] the bank raised its rates [SEP]

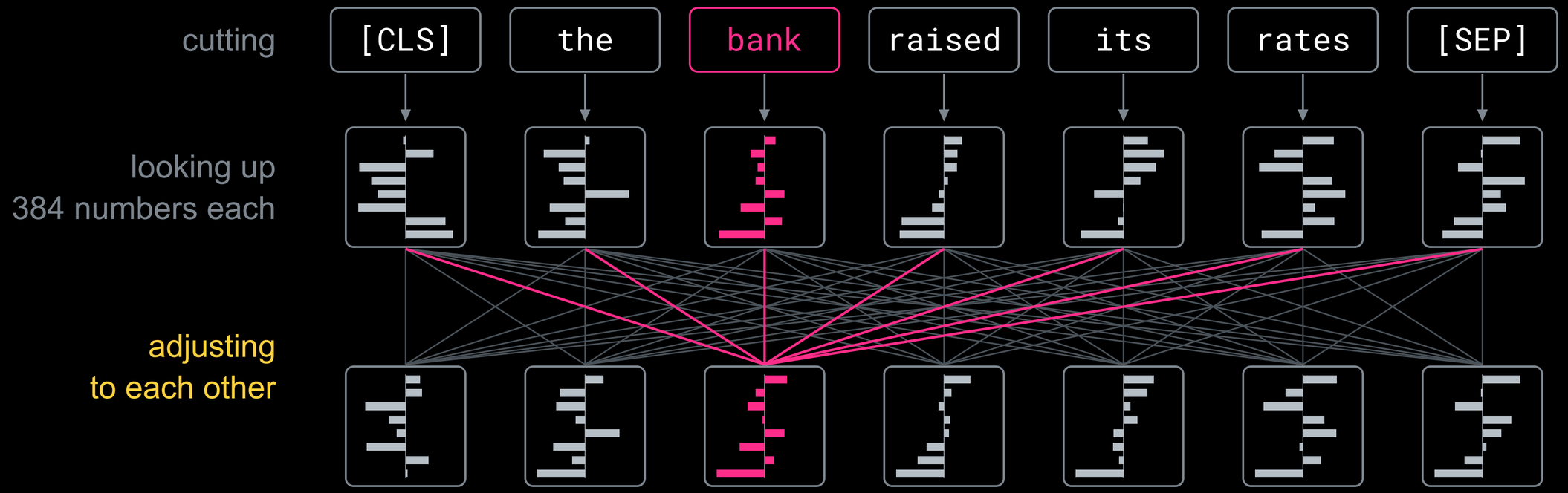
four steps from tokens to one list

cutting

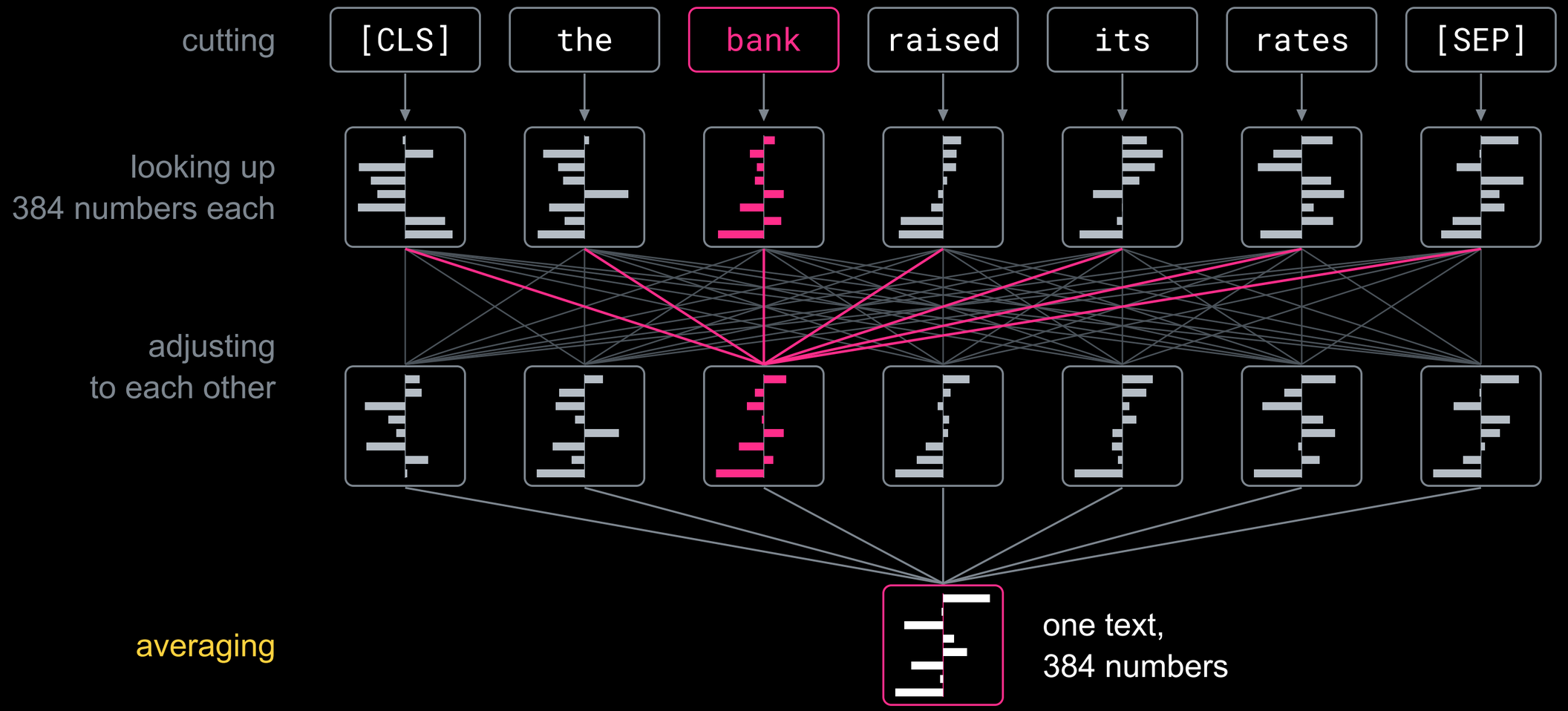
looking up
384 numbers each



four steps from tokens to one list



four steps from tokens to one list



the bank raised its interest rates.

we watched the ducks from the bank of the river.

“bank” in these two sentences	0.76
“bank” in two sentences about money	0.88

the numbers of the token after the six layers of the model

where the numbers come from

stage 1: guess the hidden word

the ____ barked
at the postman.

which word fits?

dog

puppy

words that fit the same gaps
get similar numbers

stage 2: pairs that belong together

how do i reset
my password?

closer

open settings
and click reset.

this soup needs
more salt.

further away

more than 1 billion pairs of texts

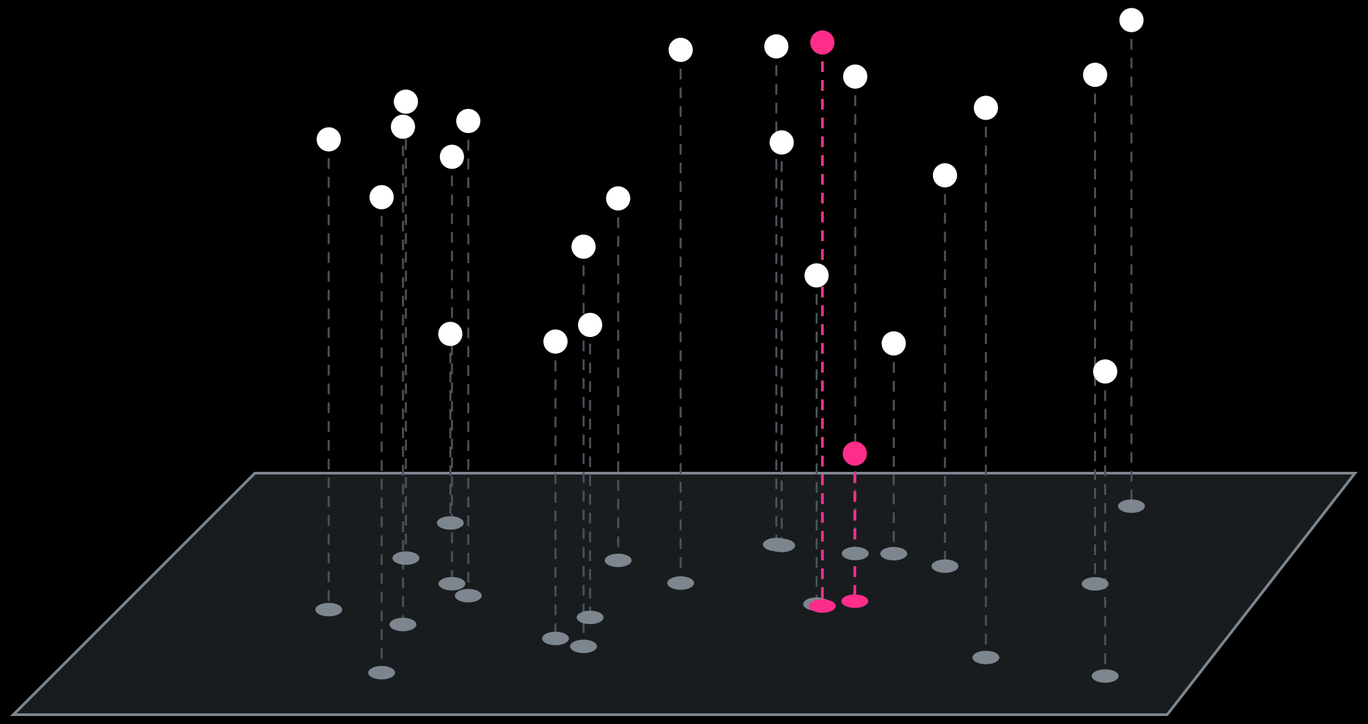
you shall know a word by the company it keeps.

John Rupert Firth, 1957

part 5

the model's map

384 directions, drawn on two



principal component analysis:
the two directions in which
the cards differ most

close in the shadow,
far apart in the numbers

★ I can't get home tonight

by meaning	1	My train was cancelled and I missed the last connection.	0.31
	2	We landed in Lisbon three hours late.	
by keywords		nothing: no card shares a word with the question	

★ I can't get home tonight

by meaning	1	My train was cancelled and I missed the last connection.	0.31
	2	We landed in Lisbon three hours late.	
by keywords		nothing: no card shares a word with the question	

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one word	`river` →	card 20 0.49	card 3 0.20
	`snake` →	card 8 0.46	card 12 0.24

a search by meaning finds texts that share no word with the question.

part 6

what the map does not know

I love this film. · I do not love this film.



0.74

I love this film. · What a wonderful movie!



0.59

I love this film. · I do not love this film.



0.74

I love this film. · What a wonderful movie!



0.59

The dog bit the man. · The man bit the dog.



0.97

Vaccines cause autism. · Vaccines do not cause autism.



0.86

a map of meaning knows what a text is about, but not whether it is true.

part 7

inside the numbers

similarity is one sum

card 1	-0.0189	-0.0445	0.0417	0.0777	0.0544	0.0546	…
card 7	0.1159	0.0211	0.0192	0.0385	0.0392	-0.0357	…

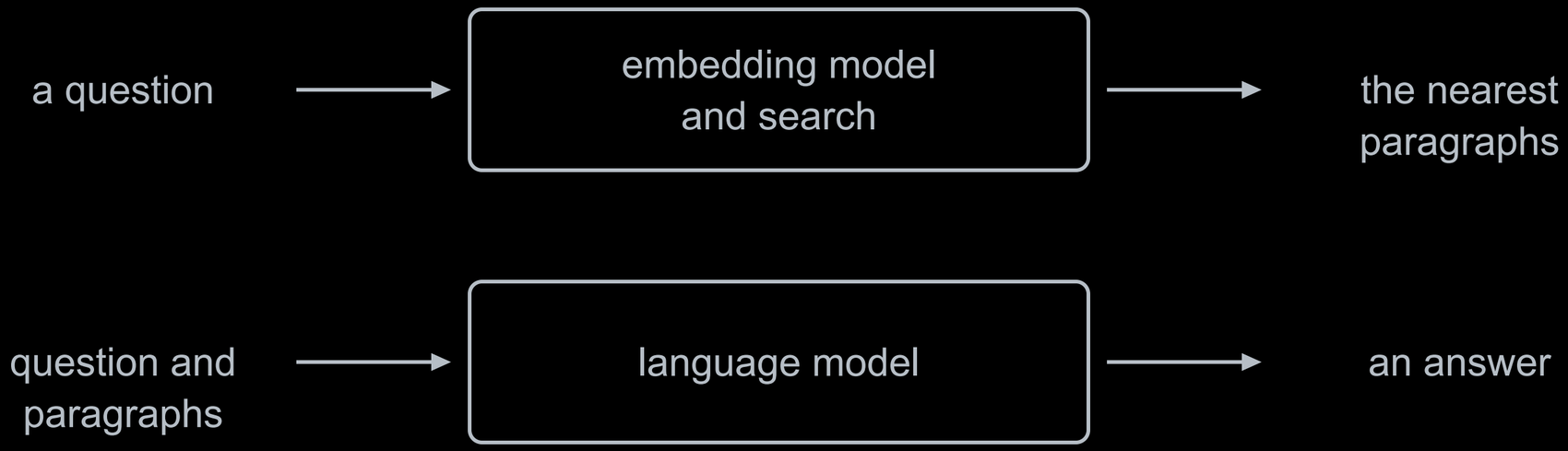
similarity is one sum

card 1	-0.0189	-0.0445	0.0417	0.0777	0.0544	0.0546	…
	×	×	×	×	×	×	
card 7	0.1159	0.0211	0.0192	0.0385	0.0392	-0.0357	…
products	-0.0022	-0.0009	0.0008	0.0030	0.0021	-0.0019	…

card 1	-0.0189	-0.0445	0.0417	0.0777	0.0544	0.0546	…
	×	×	×	×	×	×	
card 7	0.1159	0.0211	0.0192	0.0385	0.0392	-0.0357	…
products	-0.0022	-0.0009	0.0008	0.0030	0.0021	-0.0019	…

sum of all 384 = 0.32

$$\text{similarity} = a_1b_1 + a_2b_2 + \ldots + a_{384}b_{384}$$



retrieval-augmented generation

search · recommendations · duplicate questions · inside every language model